

Transitivity $\wedge$ Independence $\wedge$ Mixture continuity $\Rightarrow$ Closed-graph continuity

Result — ai; produced by Claude Fable 5.1 (Cowork session), 2026-09-08. Not yet checked.

Claim. If $\succsim$ is reflexive and transitive, satisfies independence, and has closed sections $\{\lambda : \lambda p + (1 - \lambda)q \succsim r\}$ and $\{\lambda : r \succsim \lambda p + (1 - \lambda)q\}$, then $\{(p, q) : p \succsim q\}$ is closed in $\Delta(X)^2$.

Proof. Let $V = \{x \in \mathbb{R}^X : \sum_x x = 0\}$ and $D = \Delta(X) - \Delta(X) \subseteq V$, which is compact, convex, and contains 0. Put

$$K = \{q - p : q \succsim p\}.$$

(i) *K depends only on differences.* If $q - p = q' - p'$ then $\frac{1}{2}q + \frac{1}{2}p' = \frac{1}{2}q' + \frac{1}{2}p$. By independence with $\lambda = \frac{1}{2}$, $q \succsim p \iff \frac{1}{2}q + \frac{1}{2}p' \succsim \frac{1}{2}p + \frac{1}{2}p'$ and $q' \succsim p' \iff \frac{1}{2}q' + \frac{1}{2}p \succsim \frac{1}{2}p' + \frac{1}{2}p$. The two right-hand sides are the same pair.

(ii) *K is a cone in D.* If $x \in K$, $t > 0$ and $tx \in D$, then $tx \in K$. For $t < 1$: from $q \succsim p$, independence with $r = p$ gives $tq + (1 - t)p \succsim p$, whose difference is $t(q - p)$. For $t > 1$ apply the previous case to tx .

(iii) *K is convex.* For $x = q - p$ and $y = q' - p'$ in K , independence gives $\frac{1}{2}q + \frac{1}{2}q' \succsim \frac{1}{2}p + \frac{1}{2}q'$ and $\frac{1}{2}q' + \frac{1}{2}p \succsim \frac{1}{2}p' + \frac{1}{2}p$; transitivity gives $\frac{1}{2}q + \frac{1}{2}q' \succsim \frac{1}{2}p + \frac{1}{2}p'$, with difference $\frac{1}{2}x + \frac{1}{2}y$.

(iv) *K is closed along every line.* Let $[x_0, x_1] = \ell \cap D$ and $x(\lambda) = \lambda x_1 + (1 - \lambda)x_0$. Choose $\varepsilon > 0$ and the uniform lottery c with $c + \varepsilon x(\lambda) \in \Delta(X)$ for all λ . By (ii), $x(\lambda) \in K \iff \varepsilon x(\lambda) \in K \iff c + \varepsilon x(\lambda) \succsim c$, and $c + \varepsilon x(\lambda) = \lambda(c + \varepsilon x_1) + (1 - \lambda)(c + \varepsilon x_0)$, so $\{\lambda : x(\lambda) \in K\}$ is a section, closed by mixture continuity.

(v) A convex subset of a finite-dimensional space meeting every line in a closed set is closed: for $x \in \overline{K} \setminus K$ and y in the relative interior of K , $[y, x) \subseteq K$ by the accessibility lemma, so $K \cap \text{line}(y, x)$ is not closed. Hence K is closed, and $(p_n, q_n) \rightarrow (p, q)$ with $q_n \succsim p_n$ gives $q - p \in K$. ■

Remark. This bridges the mixture-continuity axiom of the vNM literature and the closed-graph axiom of Dubra–Maccheroni–Ok. The map reaches the multi-utility representation from closed-graph continuity without it.